Quantity (বাঃ ২০১৭)

- Constant
  - ↳ a quantity which cannot be varied in time to time or situation to situation or operation to operation)
  - ↳ $y = x + 10$
    - ↳ constant
- Variable
  - ↳ a quantity which can be varied in time to time, site to site or ope to ope
  - ↳ 2 types = 1) independent v
    2) dependent v
  - ↳ Independent v ⇒ A variable which can accept or receive any value.
  - ↳ Dependent v = " " " is dependent on others.

$$\underbrace{y}_{\text{dependent}} = x + \underbrace{10}_{\text{independent}}$$

Constant &

Function

- Algebraical

d. variable

Locus of indu

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time, situ
or ope.

independent v.
dependent v.

A variable
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2 → on
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Constant & parameter

↳ a quantity which is also a kind of const constant but not absolutely constant.

$y = 2x + 10$

↳ parameter

Function

- Algebrical
- Transcendental
  - Expatention
    - $y = a^x$ (a → base, x → Index (Power)), $y = 2^x$ etc
    - a. variable
    - Index (Power)
  - logarithmic
    - ~~It should be use just for expatention~~
    - ↳ $y = a^x$
    - $x = \log_a y$
    - ↳ It depends on its base
  - Trigonometrical

Laws of indies

logarithm

Common	Natural
1) base is 10	1) base is e (e=2.718)
2) Used in Algebra	2) Used in Calculus
3) Introduced by Henry briggs	3) Introduced by John Napier

Rules of logarithm

1) $\log(mn) = \log m + \log n$

2) $\log\left(\frac{m}{n}\right) = \log m - \log n$

3) $\log x^n = n \log x$

4) $\log_a a = 1$

5) $\log 1 = 0$

6) $\log_a x = \log_b x \times \log_a b$ (Base chg formula)

Trigonometrical

Inverse

y =

Some Business / Economic

1) demand function

2) Supply "

3) Total Revenue "

4) Total Cost "

5) Total Profit "

6) Average revenue

7) " cost

8) " profit